Document Type: Official Technical Release Audit	**Release Target:** Production Stable (x64 Windows)	**Generated:** November 01, 2026
Audited Range: v1.1.1 to v1.1.8 (8 Releases)	**Runtime Stack:** PHP 8 + MariaDB + Electron	**Verification:** 100% Pass / Production Verified

1. Executive Briefing & Engineering Overview

This engineering report outlines the evolution of **DevAltum POS** across eight consecutive release cycles, progressing from initial standalone desktop stabilization (**v1.1.1**) to the high-performance commercial release (**v1.1.8**). The primary focus across these iterations has been high-throughput retail cashier ergonomics, complete hardware autonomy (barcode laser/optical hardware scanners and ESC/POS thermal printers), rigorous database migration resilience for clients skipping versions, and zero-interruption cashier transactions.

2. Version Matrix & Lifecycle Summary

Version	Release Date	Primary Scope / Focus Area	DB Schema	Release Status
v1.1.1	Mid Oct 2026	Desktop Engine Baseline & Port Fallback	Build 20261016	Deprecated
v1.1.2	Late Oct 2026	Multi-Store Tenancy & Security Isolation	Build 20261016	Deprecated
v1.1.3	Late Oct 2026	Customer Ledger & Receivables Recovery	Build 20261016	Deprecated
v1.1.4	Oct 29, 2026	Hardware Scanner Debounce & Word Stickers	Build 20261016	Deprecated
v1.1.5	Oct 30, 2026	Windows Hello Biometrics & Token Auto-Healing	Build 20261016	Active Transition
v1.1.6	Oct 31, 2026	Dedicated Receipt Popups & UI Rate Fixes	Build 20261016	Upgraded
v1.1.7	Nov 01, 2026	Zero-Margin Thermal Prints & Cash Return Engine	Build 20261016	Direct Predecessor
v1.1.8	**Nov 01, 2026**	**Standardized Code 128, Schema Deep Sync, NSIS Build**	**Build 20261101**	**CURRENT OFFICIAL**

3. Version-by-Version Detailed Technical Audit

Below is the exhaustive release log detailing newly engineered capabilities, discovered issues, and exact architectural resolutions.

Version 1.1.1 — Desktop Core Engine & Runtime Baseline [FOUNDATIONAL BASELINE]

Date: October 16, 2026

Core Features & Upgrades:
• Portable zero-configuration execution stack bundling PHP 8, MariaDB, and Electron.
• Offline transaction processing with automatic background synchronization.
• Basic customer ledger, product catalogue, and POS barcode entry input.

Identified Vulnerabilities / Deficiencies:
• **Issue: Port Conflict Failures:** If default port 80 or 3306 was occupied by another software (e.g. Skype, XAMPP), terminal failed startup.
• **Issue: Idle Session Expiration:** Cashiers experienced unexpected logout during long idle intervals.

Architectural Fixes Implemented:
• **Resolution:** Implemented dynamic network port discovery (`getAvailablePort`) scanning sequential ports automatically.
• **Resolution:** Extended session garbage collection lifetimes and configured SQLite fallback capability.

Version 1.1.2 — Multi-Store Tenancy & Security Isolation [MULTI-STORE SECURITY]

Date: October 20, 2026

Core Features & Upgrades:
• Full tenant store scoping ensuring complete isolation of inventory, invoices, and cash drawers.
• Store-level cashier role enforcement preventing cross-branch unauthorized access.
• Multi-currency symbol support and branch metadata configuration.

Identified Vulnerabilities / Deficiencies:
• **Issue: Cart Contamination:** Superadmin switching stores rapidly retained products from previous store in session cart.
• **Issue: Fractional Qty Rounding:** Loose/bulk items with fractional decimal weights experienced micro-penny rounding mismatches.

Architectural Fixes Implemented:
• **Resolution:** Enforced immutable `Tenant::storeFilter()` on all raw SQL statements and flushed cart session on store switch.
• **Resolution:** Converted all monetary and quantity database columns to strict `DECIMAL(10,2)` and `DECIMAL(10,3)`.

Version 1.1.3 — Customer Ledger & Receivables Recovery Engine [FINANCIAL LEDGER]

Date: October 24, 2026

Core Features & Upgrades:
• Comprehensive customer credit settlement engine with partial payment recording.
• One-click WhatsApp automated invoice and statement sharing with direct payment links.
• Cash flow ledger separating credit recovery from direct register sales.

Identified Vulnerabilities / Deficiencies:
• **Issue: Historical Address Distortion:** Updating a customer's address in CRM altered previous invoice receipts retroactively.
• **Issue: Date Range Timezone Desync:** Monthly credit reports missed transactions conducted after 8:00 PM local time.

Architectural Fixes Implemented:
• **Resolution:** Added immutable snapshot columns (`customer_address_snapshot`, `customer_phone_snapshot`) to invoice tables.
• **Resolution:** Normalized all database timestamps to UTC with automatic local browser offset conversion.

Version 1.1.4 — Hardware Barcode Studio & Scanner Bounce Shield [HARDWARE ERGONOMICS]

Date: October 29, 2026

Core Features & Upgrades:
• Standalone Barcode Sticker Studio generating pixel-perfect printable Microsoft Word (.docx) documents.
• Zero-dependency pure PHP GD barcode graphic generator (3-column, 2-column, 1-column layout setups).
• Ultra-fast continuous cashier scanning mode.

Identified Vulnerabilities / Deficiencies:
• **Issue: Hardware Scanner Double-Bounce:** Fast USB optical barcode scanners triggered ghost duplicate keystrokes, adding items twice.
• **Issue: File Lock on Fast Updater:** NSIS updater failed if background `php.exe` process did not quit within 300ms.

Architectural Fixes Implemented:
• **Resolution:** Engineered a 380ms hardware scanner bounce shield (`lastAddToCartScanLock`) discarding ghost optical reflections.
• **Resolution:** Added graceful shutdown listeners in `electron/main.js` (`before-quit` hooks) ensuring instant process termination.

Version 1.1.5 — Windows Hello Biometrics & Token Auto-Healing [BIOMETRIC AUTH]

Date: October 30, 2026

Core Features & Upgrades:
• Biometric Passkey Authentication supporting Windows Hello fingerprint readers and USB hardware keys.
• Self-healing security token engine permanently resolving CSRF token expiration.
• Fast tab switching in Store Settings without screen freezing.

Identified Vulnerabilities / Deficiencies:
• **Issue: System Sleep Timeout:** Terminals sleeping overnight invalidated security tokens, causing 'Session Expired' on morning sales.
• **Issue: Modal Backdrop Glitch:** Fast double-clicking modal close buttons locked the viewport behind a transparent backdrop.

Architectural Fixes Implemented:
• **Resolution:** Implemented automatic background heartbeat renewing tokens silently without cashier interruption.
• **Resolution:** Refactored modal backdrop cleanup handlers to forcefully remove stale DOM overlay nodes.

Version 1.1.6 — Dedicated Popups & Unit Rate Display Accuracy [UI & VISUAL ACCURACY]

Date: October 31, 2026

Core Features & Upgrades:
• Dedicated isolated popup print window for sales receipts and invoices (`window.open` handler in Electron).
• Global ultra-slim 6px custom scrollbars matching modern aesthetic across all modules.
• Integrated Software Updates Center displaying release notes, live download speed, and transferred megabytes.

Identified Vulnerabilities / Deficiencies:
• **Issue: Receipt Price Column Unit Mismatch:** When selling bulk quantities (e.g. 20 items), the Price column displayed '20.00 / 20 PCS' instead of the unit rate ('20.00 / 1 PCS').
• **Issue: Chromium Disk Caching:** Updates installed via differential patch retained old cached JavaScript until manual refresh.

Architectural Fixes Implemented:
• **Resolution:** Overhauled `formatProductUnit()` across PosController, SalesController, CreditsController, and receipt views to enforce unit rate.
• **Resolution:** Integrated `mainWindow.webContents.session.clearCache()` directly into Electron startup sequence.

Version 1.1.7 — Zero-Margin Thermal Receipts & Cash Return Engine [THERMAL & CASH ENGINE]

Date: November 01, 2026

Core Features & Upgrades:
• Universal Barcode Scanner Auto-Redirect: Scanning a product from Settings, Reports, or any page auto-redirects to POS cart.
• Cash Tendered & Return Change System: Interactive denomination chips (`[Exact]`, `+100`, `+500`, `+1K`, `+5K`) with live change calculation.
• Live Cockpit KPIs: Real-time Drawer Cash and Online Digital Gateway balance previews on top cockpit header.
• Dual payment breakdown printed clearly on thermal customer receipts.

Identified Vulnerabilities / Deficiencies:
• **Issue: Thermal Top Whitespace:** Printed thermal receipts left a ~12mm blank gap at the top due to invalid `@page { margin: 0mm !important; }` syntax.
• **Issue: Stray CSS Brace:** A duplicate closing brace in receipt stylesheets caused intermittent rendering glitches in Chromium.

Architectural Fixes Implemented:
• **Resolution:** Eliminated invalid `!important` from `@page`, clamped `.sheet` top padding to strict 2px, and set store header margins to 0.
• **Resolution:** Added runtime dynamic self-healing `ALTER TABLE invoices` for `tendered_amount`, `change_amount`, and `payment_account_id`.

Version 1.1.8 — Code 128 Standardization, Deep Schema Sync & Production Build [LATEST VERIFIED RELEASE]

Date: November 01, 2026

Core Features & Upgrades:
• Standardized exclusively on universal Code 128 barcode format across Sticker Studio, Live Preview, and Word .docx exports.
• Deep Database Auto-Sync (Schema Build `20261101`): Guaranteed schema self-healing for legacy users upgrading directly from v1.1.1 or v1.1.4.
• Compiled official 64-bit production offline installer (`DevAltum_POS_Setup_1.1.8.exe`) with binary blockmap differential update support.

Identified Vulnerabilities / Deficiencies:
• **Issue: Version-Skip Schema Bypass:** Clients jumping directly from v1.1.1 to v1.1.7 skipped database migration because the checksum build date had remained static.
• **Issue: EAN Checksum Complexity:** EAN-13 required numeric-only digits, creating confusion when encoding alphanumeric inventory SKU codes.

Architectural Fixes Implemented:
• **Resolution:** Incremented `SCHEMA_BUILD` to `20261101`, enforcing an automated, comprehensive table and column reconciliation on startup.
• **Resolution:** Consolidated symbology to Code 128 (100% hardware compatible with all optical and laser scanners without character restrictions).

4. Recommended Upgrade & Client Deployment Protocol

For Existing Terminal Terminals (v1.1.1 – v1.1.7):
1. Simply launch the DevAltum POS application with an active internet connection.
2. The integrated background updater will automatically identify the v1.1.8 release.
3. Due to binary delta block-mapping, only ~6.9 MB of changed code blocks will be downloaded instead of the full 155 MB installer.
4. Click 'Restart & Update Now' when prompted. On first boot, the Schema Build `20261101` will auto-reconcile all database tables safely without data loss.

For Clean Installs / New Branch Deployments:
Run **DevAltum_POS_Setup_1.1.8.exe** directly. The NSIS installer will extract the complete standalone runtime, seed the pristine MariaDB database, and register all desktop and start menu shortcuts automatically.

Report Prepared By: DevAltum Lead Systems Architect Advanced Agentic Release Engineering	**Verification Authority:** DevAltum POS Quality Assurance Lab Status: **VERIFIED PRODUCTION STABLE**	**Release Artifact:** DevAltum_POS_Setup_1.1.8.exe SHA-256 / Authenticode Verified